DHRUVA

Voice-first marine decision support for India's small-scale fishermen

Smart India Hackathon 2026 · ISRO problem statement **SIH26176**: ORCA, Marine EcOsystem Reasoning with Collaborative Agents · Bhashini API key request for app **KEY_1_DHRUVA**

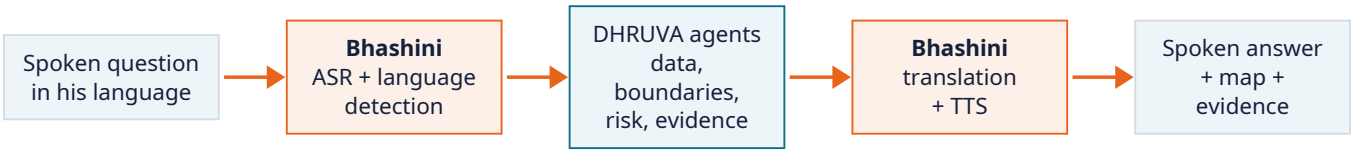
The problem

ISRO and INCOIS publish rich ocean data every day: sea surface temperature, chlorophyll, waves, wind and potential fishing zones. It reaches fishermen as coast-wide bulletins, mostly as text. A fisherman who reads little, or owns only a feature phone, cannot ask the question he actually has: *is it safe for my boat, here, this morning, and where should I go?*

What DHRUVA does

He asks out loud in his own language. A team of AI agents works out what he is asking, fetches the satellite and ocean data for his location, checks maritime boundaries, and gives a **go / caution / no-go** answer with the reasons. The answer is spoken back in the same language, with a map and an evidence card showing which dataset and time every figure came from. A validator rejects any number in the answer that is not in the fetched data, so the AI cannot invent a wave height.

Where Bhashini fits



Services requested

Service	Why DHRUVA needs it
ASR (speech to text)	The fisherman speaks his question instead of typing it.
Language detection	Reply in the language he asked in, a stated requirement of SIH26176.
Translation (NMT)	Bridge between his language and the English the agents reason in.
TTS (text to speech)	Speak the answer for users who do not read, on app and phone line.

Languages

Tamil, Malayalam, Telugu, Kannada, Marathi, Konkani, Gujarati, Odia, Bengali, Hindi, Urdu and English, covering India's coastal states. Answer templates already exist in all twelve.

Channels

Web app (PWA) for smartphones, including low-end Android, plus a shore console for fisheries officers. **IVR phone line** so a feature-phone user can call, ask and hear the answer.

Use of the key

All Bhashini calls will be made from our backend server; the key is never shipped to a user's device. Every answer carries the note *advisory only, follow official INCOIS and IMD warnings*. Non-commercial student prototype.

Status

Built	Agent backend (FastAPI, LangGraph) with a Gemini planner; live ocean data from Copernicus Marine and Open-Meteo; INCOIS potential-fishing-zone scraper; maritime boundaries in PostGIS; risk rules and the number validator; web app with maps and satellite layers; answer templates in twelve languages.
Next	Voice in and out through Bhashini (this request), then the IVR line.